

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 7: Shared Decisions, Hormonal Health and Biotechnology: Our Final Explanation of How Animals Reproduce

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	A	A	A	D	D	A

USE AFTER: PREDICT PHASE — Predict: sticky-note prediction on why frogs lay thousands of eggs but a cow has one calf

Q1. A frog near Kisumu lays thousands of eggs, but a cow gives birth to only one calf at a time. Which statement best explains this difference?

- A. Frog eggs have hard shells, so many of them are needed to break through
- B. Frogs are smaller than cows, so their bodies are able to make more eggs than a cow can
- C. Frogs fertilise eggs in water with no parental care, so many eggs are lost to predators and the environment, while a cow's young develop inside the body and are cared for**
- D. Cows produce hormones and frogs do not, so cows cannot make many young

Correct answer: C. External fertilisation with no parental care means high losses, so many eggs are needed. Internal development and care in cows protect the young, so one calf is enough.

USE AFTER: OBSERVE PHASE — Station 1: small mammal specimen and labelled chart of the male and female reproductive systems

Q2. The uterus of a mammal has a thick, muscular wall. Which function does this structure fit best?

- A. It holds and protects the developing embryo and contracts to push the baby out at birth**
- B. It carries sperm from the testes to the outside of the body
- C. It produces eggs and releases them every month
- D. It produces sperm cells for fertilisation

Correct answer: A. The uterus is where the embryo develops. Its thick muscular wall protects the embryo and contracts strongly during birth.

USE AFTER: OBSERVE PHASE — Station 2: classification chart (oviparous, viviparous, ovoviviparous)

Q3. In a certain snake, the eggs are fertilised inside the female. The eggs stay and hatch inside her body, and the young are born alive. The young feed on egg yolk, not through a placenta. How is this snake classified?

- A. Ovoviviparous, because eggs develop and hatch inside the mother's body**
- B. An external fertiliser, because it produces eggs
- C. Viviparous, because the young are born alive
- D. Oviparous, because it makes eggs

Correct answer: A. In ovoviviparous animals, eggs develop and hatch inside the mother and the young feed on yolk. Viviparous animals nourish young through a placenta, and oviparous animals lay eggs that develop outside the body.

USE AFTER: OBSERVE PHASE — Station 2: animation clip on fertilisation, implantation, placenta and birth

Q4. In a human pregnancy, the placenta joins the mother's body to the developing baby. What happens across the placenta?

- A. Oxygen and nutrients pass from the mother's blood to the baby, and wastes pass from the baby to the mother's blood**
- B. The mother's blood and the baby's blood mix completely and flow in one loop
- C. Nothing passes across it, because the baby feeds only on a yolk store
- D. Only milk passes from the mother to the baby

Correct answer: A. The placenta allows exchange of oxygen, nutrients and wastes between the two blood supplies without the bloods mixing. Human embryos have no large yolk store and depend on this exchange.

USE AFTER: OBSERVE PHASE — Debate preparation: hormone flowcharts (FSH, LH, oestrogen, progesterone, oxytocin, prolactin, testosterone) for Group D

Q5. Which hormone stimulates the muscles of the uterus to contract strongly during childbirth?

- A. Testosterone
- B. Prolactin
- C. FSH
- D. Oxytocin**

Correct answer: D. Oxytocin causes the uterine muscles to contract during birth. Prolactin stimulates milk production, FSH helps eggs to mature, and testosterone is the main male sex hormone.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning Final Explanation, part 3 on IVF, surrogacy and sperm banks

Q6. A news report says a Kenyan couple had a baby through IVF. Which statement correctly describes where the baby grew?

- A. The baby grew in an incubator in the hospital after the embryo was made
- B. The baby grew wholly in a test tube until birth
- C. Fertilisation happened in the uterus, then the embryo was moved to a test tube to develop
- D. Fertilisation happened outside the body, then the embryo was placed in a uterus where it developed until birth**

Correct answer: D. In IVF only fertilisation and early cell division happen outside the body. The embryo then develops in the uterus of the mother or a surrogate, with a placenta, until birth.

USE AFTER: MODEL BUILDING PHASE — Model: section 3 of the final synthesis model on IVF, surrogacy and sperm banks

Q7. A woman's ovaries make healthy eggs, but her uterus is damaged and cannot carry a pregnancy. Which technology would let the couple have their own biological child?

- A. Surrogacy, because another woman's uterus can carry the couple's embryo made by IVF**
- B. A sperm bank, because it supplies donated sperm cells
- C. External fertilisation in water, as in frogs
- D. Growing the embryo wholly in a test tube until the baby is ready to be born

Correct answer: A. The problem is the uterus, not the eggs or sperm. Surrogacy with an embryo made by IVF lets another woman's uterus carry the baby. A sperm bank only supplies sperm, and an embryo cannot develop wholly in a test tube.